

## Final Project Proposal Assignment

Due Sunday 10/25/26 by midnight. No late work will be accepted, assignment is required to pass the class.

The big goal of this class is to learn programming for scientific applications. To this end, a good chunk of the semester will be spent working on an individual research project using scientific data analysis. Students are encouraged to do a project related to their academic research in progress, or topics they wish to explore in the future. In previous classes students have used the final project to work on methods for their theses, but basically, you can do whatever you are into. It is critical here that you pick a **small, tractable** project that is easy to do in a few weeks of working 10-12 hours per week on the programming and analysis, plus time for the write up.

Learning outcomes:

The goal of this assignment is to familiarize you with scientific proposal structure and writing process, get you planning for your final project, and have you gain familiarity in integrating Zotero, python output and text editors. In life, proposals have hard deadlines. This assignment has a hard deadline, and late work will not be accepted.

Assignment:

In Word or the LaTeX editor of your choice, write a short, 2–4 page research proposal, including: abstract, background, why it's important, what your hypothesis is, what analysis you plan to do, what the broader impacts are, and a timeline for completion. (Broader impacts: How can this information be used? Why do we care? Impact for society...) Two figures (at least one that you created, i.e. preliminary results) and five references (using Zotero or similar) are required for grad students. Put the most energy into thinking and writing about what your planned analysis is, how it will answer your research questions, and how you will accomplish this with the time you have. **DO NOT PUT TOO MUCH TIME OR TEXT INTO THE BACKGROUND**; it is suggested you write this last.

You will also give a short class presentation to get feedback on your proposed research on 10/26.

Common issues from previous semesters:

- Put name in both name of file and in the document on the first page or as a header
- Use standard paragraph format: indent or space
- Figures need to have captions that explain what they are
- Every visual element (figures, tables, timelines) needs to be cited in the main text and narrated as to what they are included for. They need context.
- Research questions go before the methods. In the methods you explain how you will answer the questions explicitly.

- If you introduce indices/metrics, you need to define them in the text and provide a citation
- The timeline needs to be narrated in text form (there can also be an image/table) and integrated into the methods
- Broader impacts should not be longer than the methods
- Figure captions need to be on the same page as a figure.
- Section headings should not be at the end of the page, insert a page break
- Text is way too detailed for the assignment and reads like it was copied from another document/proposal without being appropriately edited down.
  - Abstract reads like the proposed research was already completed
  - Too much detail on metrics and not on what programming will be done

Save this as a pdf and load into github as final\_proposal\_yourname.pdf. Include a .zip file of all of your overleaf source files or a screenshot of your bibliography manager if using word.

As per the syllabus timely and satisfactory completion of this assignment is needed to pass the class. Late submissions will not be accepted except in extreme circumstances and with prior approval and may result in an incomplete or failing grade given in the course.